CSE 2218: Data Structure and Algorithms II Laboratory

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Recursion

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Definition

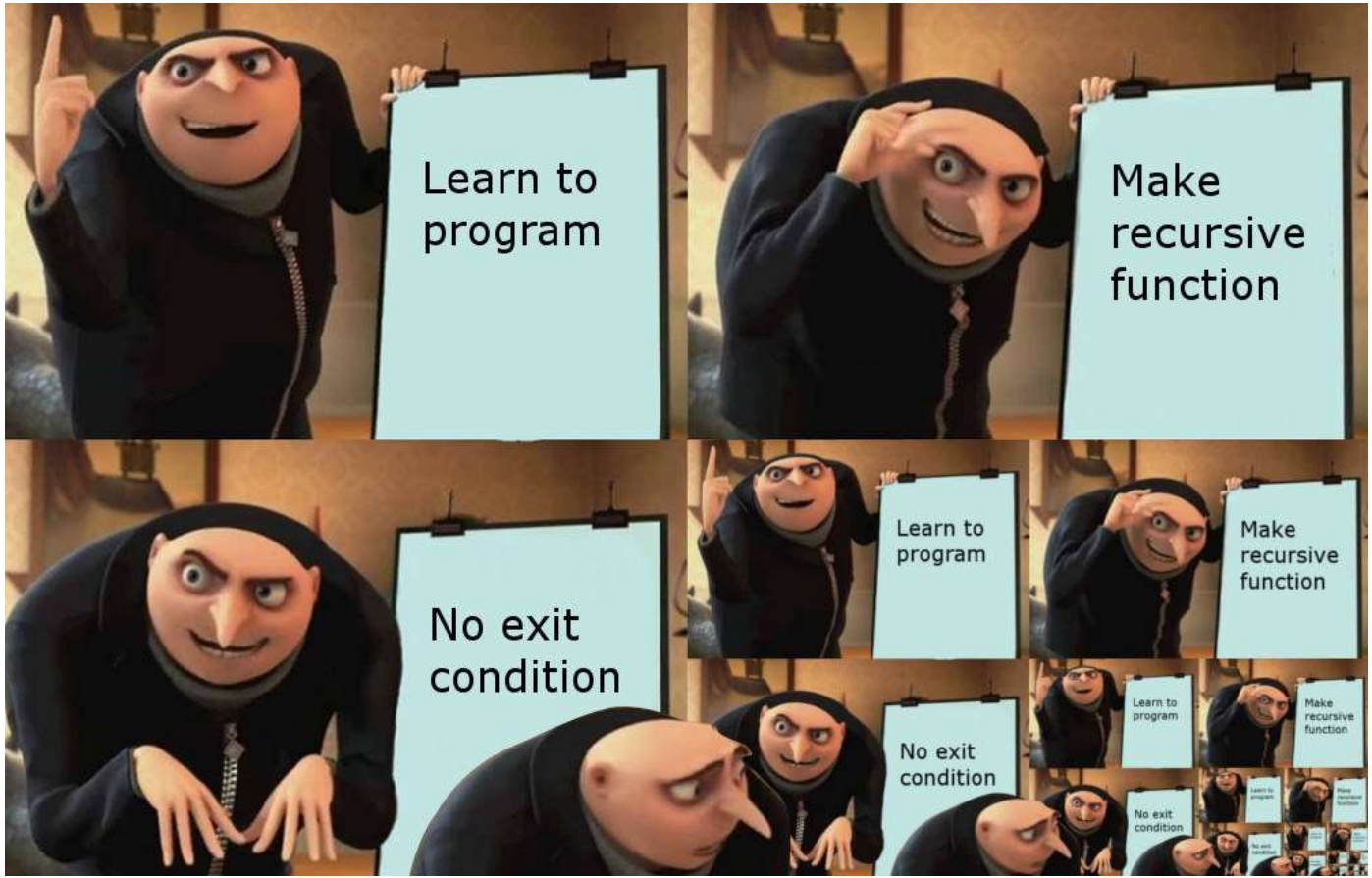
A function that calls itself to solve a problem

Two main components:

- **Base** Case: The stopping condition.
- **Recursive** Case: The function calls itself with modified parameters.

Why Use Recursion?

- Simplifies complex problems.
- Natural fit for problems like divide and conquer.
- Reduces the need for explicit loops and complex iterations.


Learn to program
Make recursive function
No exit condition
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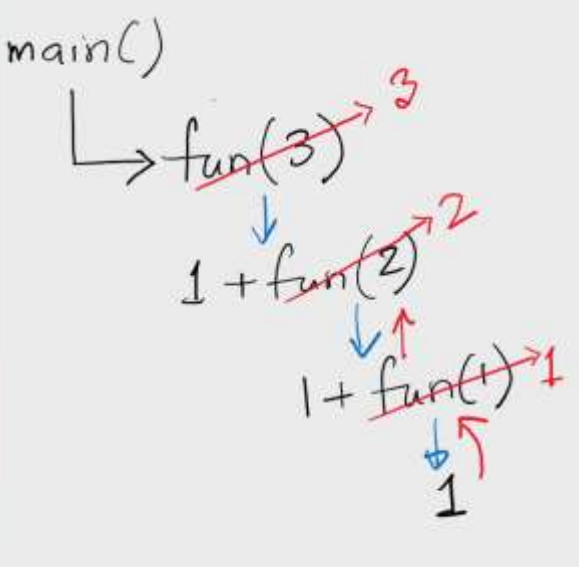
Simulation of a recursive function:

```
int fun(int n)
{
    if(n == 1)
        return 1;
    else
        return 1 + fun(n-1)
}

int main() {
    int n = 3;
    printf("%d", fun(n));
    return 0;
}
```



How to write recursive functions

1. Divide the problem into smaller subproblems. Identify the recursive formula.

```
   Fact (1) = 1
Fact (2) = 2. 1 = 2. Fact (1)
Fact (3) = 3. 2. 1 = 3. Fact (2)
Fact (4) = 4. 3. 2. 1 = 4. Fact (3)
…
Fact (n) = n. Fact(n-1)
```

How to write recursive functions

2. Specify the base condition to stop the recursion

The base case returns a value without making any subsequent recursive calls. For example: Fact (1) = 1

```
Function rec:
  if base condition:
  ...
  else
  ... recursive formula
```

```
Function Fact(n):
  if n==1 then
    return 1
  else
    return n*Fact(n-1)
```

Factorial Calculation

Write a recursive implementation of the factorial function.

Recall that $n! = 1 \times 2 \times \ldots \times n$, with the special case that $0! = 1$

Factorial Calculation

```
#include <iostream>
using namespace std;

int factorial(int n) {
    if (n == 0)  // Base case
        return 1;
    return n * factorial(n - 1);  // Recursive case
}

int main() {
    cout << factorial(5);
    return 0;
}
```

Fibonacci

Write a recursive program to print the nth Fibonacci number.

Fibonacci

```
#include <iostream>
using namespace std;

int fibonacci(int n) {
    if (n <= 1)  // Base case
        return n;
    return fibonacci(n-1) + fibonacci(n-2);  // Recursive case
}

int main() {
    cout << fibonacci(6);
    return 0;
}
```

Sum of Digits

You are given an integer number. You need to calculate the sum of its digits using recursion.

Sum of Digits

```
#include <iostream>
using namespace std;

int sumDigits(int n) {
    if (n == 0)  // Base case
        return 0;
    return n % 10 + sumDigits(n / 10);  // Recursive case
}

int main() {
    cout << sumDigits(1234);
    return 0;
}
```

Reverse a string

Write a recursive function to reverse a string.

Reverse a string

```
#include <iostream>
using namespace std;

string reverseString(const string &str, int index) {
    if (index < 0)
        return "";
    return str[index] + reverseString(str, index - 1);
}

int main() {
    string str = "hello";
    string reversed = reverseString(str, str.length() - 1);
    cout << "Reversed string: " << reversed << endl;

    return 0;
}
```

Task

1. Write a recursive program to count the number of digits of an integer.
2. Write a recursive program to calculate the power of x (x^y), where y is a non-negative integer.
3. Write a recursive program to check if a given string is a palindrome or not (not case sensitive, ignore whitespaces)

Task

4. Write a recursive implementation of binary search in a sorted array.

5. Write a recursive program to find the GCD of x and y where x, y are positive integers. (Hint: use Euclid's algorithm)

6. Write a recursive program to print an array of size n in reverse order.

Task

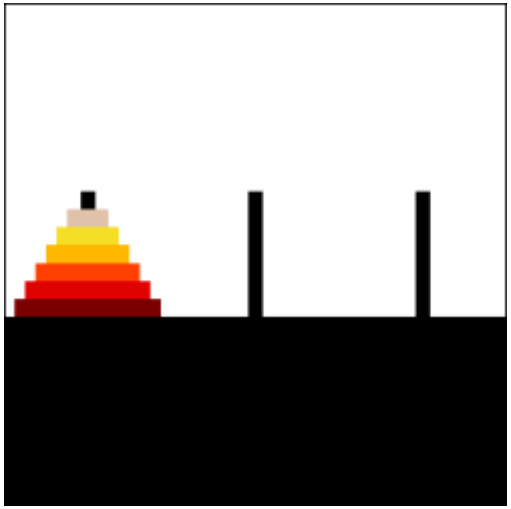
7. Find the sum of the following series up to n^{th} position: $1^2+2^2+3^2+\ldots$

8. Write a recursive program to check if a given positive integer is a palindrome or not. An integer is a palindrome when it reads the same backward as forward.

9. Write a recursive program to print all subsets of a set of n elements.

Task

10. Write a recursive program to solve the Tower of Hanoi problem for n disks.



Sample input	Sample output
4	Move disk 1 from A to B Move disk 2 from A to C Move disk 1 from B to C Move disk 3 from A to B Move disk 1 from C to A Move disk 2 from C to B Move disk 1 from A to B Move disk 4 from A to C Move disk 1 from B to C Move disk 2 from B to A Move disk 1 from C to A Move disk 3 from B to C Move disk 1 from A to B Move disk 2 from A to C Move disk 1 from B to C

Thank you!